

CO2 - Assment 601 I

1. Suppose that the data for analysis includes the attributes age. The age values for the data tuples are (increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

(a) what is the mean of the data? what is the median?

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad \mu = \frac{\sum x}{N}$$

the sum of all 27 age values is 809

$$\text{mean} = \frac{809}{27} \approx 29.96$$

$$\text{median} = 25 \quad \left(\because \frac{27+1}{2} = 14^{\text{th}} \right)$$

(b) what is the mode of the data? comment on the data's modality (i.e. bimodal, trimodal, etc.)

frequency breakdown:

25 appears 4 times

35 appears 4 times

Modality: The dataset is bimodal
has exactly two modes

(c) what is the midrange of the data?

minimum value (min): 13

maximum value (max): 70

$$\text{midrange} = \frac{\text{min} + \text{max}}{2} = \frac{13 + 70}{2} = \frac{83}{2} = 41.5$$

(d) can you find (roughly) the first quartile (Q_1) and the third quartile (Q_3) of the data?

1. 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25
first quartile
 $Q_1 = 20$

2. 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70
 $Q_3 = 35$

3. give the five-number summary of the data

min - 13

Q_1 - 20

median - 25

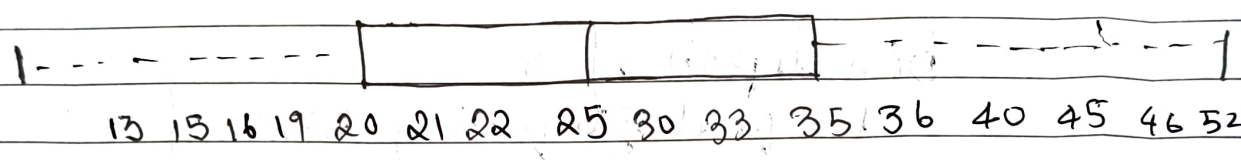
Q_3 - 35

max - 70

(13, 20, 25, 35, 70)

(f) Show a boxplot of the data

Box: extends from $Q_1 = 20$, $Q_3 = 35$
median line at 25
outlier: 70



2. Given the following measurements for the variable age:

18, 22, 25, 42, 28, 43, 33, 35, 56, 28, standardize the variable by the following:

Compute the mean absolute deviation of age.

$$\bar{x} = \frac{330}{10} = 33$$

Age	$ x_i - \bar{x} $	
18	$18 - 33 = 15$	Sum of absolute deviations $= 15 + 11 + 8 + 9 + 5 + 10 + 0 + 2 + 23 + 5 = 88$ $= \frac{88}{10} = 8.8$
22	$22 - 33 = 11$	
25	$25 - 33 = 8$	
42	$42 - 33 = 9$	
28	$28 - 33 = 5$	
43	$43 - 33 = 10$	
33	$33 - 33 = 0$	
35	$35 - 33 = 2$	
56	$56 - 33 = 23$	
28	$28 - 33 = 5$	

Q3 Suppose that the data for analysis includes the age attribute. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. Use smoothing by bin means to smooth the above data, using a bin depth of 3. illustrate your steps.

=> bin depth of 3
9 equal frequency bins of 3 items each

Bin	Original values	Mean	Smoothed values
B ₁	13, 15, 16	$(13+15+16)/3 = 14.67$	14.67, 14.67, 14.67
B ₂	16, 19, 20	18.33	18.33, 18.33, 18.33
B ₃	20, 21, 22	21.00	21, 21, 21
B ₄	22, 25, 25	24.00	24, 24, 24
B ₅	25, 25, 30	26.67	26.67, 26.67, 26.67
B ₆	33, 33, 35	33.67	33.67, 33.67, 33.67
B ₇	35, 35, 35	35.00	35, 35, 35
B ₈	36, 40, 45	40.33	40.33, 40.33, 40.33
B ₉	46, 52, 70	56.00	56, 56, 56

~~B₁₀~~

~~B₁₁~~

~~B₁₂~~

- Q4 use the two methods below to normalize the following group of data: 200, 300, 400, 600, 1000
 (a) min max normalization by setting min=0 and max=1

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

min value = 200

max value = 1000

$$\max - \min = 800$$

original value calculation
 (x)

200	$(200 - 200 / 1000 - 200) = 0$
300	$(300 - 200) / (1000 - 200) = 0.125$
400	$(400 - 200) / (1000 - 200) = 0.25$
600	$(600 - 200) / (1000 - 200) = 0.5$
1000	$(1000 - 200) / (1000 - 200) = 1$

0, 0.125, 0.25, 0.5

- (b) ~~Suppose a group of 12 sales price records has been sorted as follows~~

(b) Z-score normalization

$$V' = \frac{V - \bar{x}}{\sigma}$$

mean $\bar{x} = 500$

(c)

$$(200 - 500)^2 = 90,000$$

$$(300 - 500)^2 = 40,000$$

$$(400 - 500)^2 = 10,000$$

$$(600 - 500)^2 = 10,000$$

$$(1000 - 500)^2 = 250,000$$

$$\text{Sum} = 400,000$$

$$\sigma = \sqrt{\frac{400,000}{5}} = 282.84$$